

TruthLens 499B: Evidence-Grounded Multimodal Misinformation Verification for Bangla Social-Media Content

Interim Research Progress Report

Anas Ahmed
ID: 2121264042
anas.ahmed@northsouth.edu

Subroto Ghosh
ID: 2212150642
subroto.ghosh@northsouth.edu

Joy Saha
ID: 2312466042
joy.saha1@northsouth.edu

Md. Abu Sufian Protik
ID: 2312234042
abu.protik@northsouth.edu

Faculty Advisor

Dr. Mohammad Shifat-E-Rabbi (MSRB)

Assistant Professor, Department of Computer Science and Engineering
North South University, Dhaka, Bangladesh
rabbi.mohammad@northsouth.edu

Abstract—Bangla misinformation circulates through text, images, and videos, yet conventional fake/real classifiers do not necessarily identify the specific claim being judged or provide verifiable supporting evidence. TruthLens 499B investigates evidence-grounded multimodal verification of Bangla social-media content. The proposed workflow extracts checkable claims, retrieves external evidence, preserves source provenance, and assigns one of three verdicts: Supported, Refuted, or Insufficient. Phase 1, comprising the dataset schema and annotation guidelines, is complete. Phase 2, a pilot targeting approximately 100 samples, is in progress. No pilot statistics or model results are available yet. This report summarizes the motivation, research questions, current progress, proposed evaluation, and remaining work.

Index Terms—Bangla misinformation, multimodal verification, evidence retrieval, fact-checking, data leakage

I. INTRODUCTION

Misinformation shared through Bangla social media can be difficult to verify at scale. Existing resources, including BD-FakeDetect, MultiBanFakeDetect, and BTMD, provide useful foundations for Bangla misinformation detection across different data types [1]–[3]. However, predicting whether a post is fake or real does not necessarily establish which factual claim is being evaluated or whether reliable evidence supports the prediction.

Evidence-based benchmarks such as FEVER and AVeriTeC demonstrate the importance of claim-level verification, explicit verdicts, and evidence-based justification [4], [5]. Multimodal verification research also highlights the importance of checking relationships between media and contextual evidence [6], [7].

TruthLens 499B explores a claim-centric approach for Bangla social-media content, particularly video. Its potential contribution is not simply combining Bangla, video, or language models, but investigating evidence provenance, evidence sufficiency, human-verified verdicts, and leakage-resistant evaluation together. This contribution remains a research hypothesis to be tested.

II. RESEARCH GAP AND QUESTIONS

The project investigates four candidate gaps:

- **Evidence sufficiency:** Retrieved information may be relevant without adequately supporting or refuting a claim.
- **Cross-modal inconsistency:** Captions, video frames, audio, and external reports may refer to different events, dates, or locations.
- **Missing modalities:** Verification may become less reliable when video, audio, or text is unavailable.
- **Data leakage:** Duplicate media, repeated claims, or posts about the same event may appear across training and test sets, inflating evaluation scores.

These gaps motivate the following research questions:

- 1) Does evidence-grounded verification provide more verifiable explanations than classification-only baselines?
- 2) Can annotators consistently distinguish Supported, Refuted, and Insufficient claims?
- 3) How frequently do media content and retrieved evidence disagree in context?
- 4) Do event-, claim-, and media-grouped splits produce different results from random splits?

These questions may be narrowed after the pilot and further literature review.

III. PROPOSED METHODOLOGY

The planned pipeline is:

Bangla Content → **Preprocessing** → **Claim Extraction** →
Evidence Retrieval → **Verification** → **Explanation**

For videos, preprocessing may include frame sampling, speech transcription, and optical character recognition (OCR). Checkable claims will be extracted and matched against candidate evidence from sources such as fact-checking organizations, reputable news outlets, official documents, and relevant web pages. Each evidence item should retain its URL, publication date where available, and supporting passage.

The proposed verdicts are:

TABLE I
PROJECT STATUS AND NEXT ACTIONS

Activity	Status	Next Action
Dataset schema	Completed	Apply schema to pilot samples
Annotation guidelines	Completed	Validate consistency on real samples
Pilot study (≈ 100 samples)	In progress	Assess collection and annotation feasibility
Evidence retrieval pipeline	Planned / not confirmed	Test retrieval and provenance recording
Larger dataset	Planned	Scale only if the pilot justifies it
Model evaluation	Planned	Establish baselines and grouped splits

- **Supported:** Reliable evidence adequately supports the claim.
- **Refuted:** Reliable evidence adequately contradicts the claim.
- **Insufficient:** Available evidence does not justify either conclusion.

Human annotators will determine the reference verdict using original sources, reliable reporting, fact-checks, and contextual information. AI tools may assist with collection, transcription, OCR, duplicate detection, claim extraction, and evidence discovery, but their outputs will not be treated as ground truth without human verification.

IV. PROGRESS TO DATE

A. Phase 1: Dataset Schema and Guidelines

Phase 1 is complete according to the project team. The planned schema organizes information into sample identity, original content, extracted claims, evidence, human judgments, and evaluation-group identifiers. These fields are intended to support traceability, reproducible annotation, and leakage-resistant evaluation.

Completion of the schema and guidelines does not establish that they have been validated, that the complete dataset exists, or that the verification pipeline has been implemented.

B. Phase 2: Pilot Study

Phase 2 has started, with a target of approximately 100 samples. The pilot is intended to test whether Bangla content can be collected consistently, claims can be extracted, reliable external evidence can be found, and annotators can apply the verdict labels consistently.

The pilot will also examine duplicate content, shared events, missing media, annotation effort, and disagreements. A subset should be independently annotated, with disagreements reviewed through an explicit adjudication process. Agreement statistics, collection counts, retrieval success rates, and annotation-time measurements will be reported only after they have actually been measured.

V. EVALUATION AND DATA QUALITY

The pilot will guide decisions about whether to revise the guidelines, narrow the scope, or scale the dataset. Planned measurements include:

- **Annotation reliability:** Cohen’s or Fleiss’ kappa, as appropriate, alongside disagreement analysis.
- **Verification quality:** Macro-F1, per-class precision and recall, and confusion matrices.
- **Evidence quality:** Retrieval relevance, sufficiency, source reliability, and provenance completeness.
- **Robustness:** Performance when individual modalities are unavailable.
- **Leakage checks:** Exact and near-duplicate media, related claims, shared events, and source overlap across partitions.

Random splitting alone may place closely related samples in different partitions. The proposed evaluation will therefore group related claims, events, and media before assigning train, validation, and test partitions. Random-split results may be retained as a comparison, not as the only evidence of generalization.

Candidate baselines include a majority-class predictor, a text-only classifier, a multimodal classifier without retrieval, and an evidence-aware system. Comparisons will use consistent samples, labels, and evaluation partitions. These experiments remain planned; no model has yet been reported as evaluated.

VI. LIMITATIONS AND REMAINING WORK

The main anticipated challenges are limited annotation capacity, ambiguous claims, unavailable or unreliable external evidence, poor speech transcription or OCR, platform access restrictions, class imbalance, and duplicate or event-related overlap. Data rights, privacy, and permitted media redistribution must also be considered before releasing a dataset.

The remaining work is organized into four stages:

- 1) **Complete the pilot:** Document sample counts, annotation disagreements, evidence availability, and effort.
- 2) **Build the pipeline:** Implement and test retrieval, provenance recording, and annotation support.
- 3) **Scale if justified:** Consider a dataset of 1,000–1,500 samples only if pilot feasibility, annotation quality, and available resources support it.
- 4) **Evaluate and analyze:** Run baselines, grouped-split experiments, modality ablations, and error analysis.

VII. CONCLUSION

TruthLens 499B aims to investigate evidence-grounded multimodal misinformation verification for Bangla social-media content. Phase 1, covering the dataset schema and annotation guidelines, is complete, while the approximately 100-sample Phase 2 pilot is in progress. The central next milestone is a documented pilot assessment establishing whether the annotation process is reliable, sufficient evidence is available, and the proposed dataset can be collected within the project’s

constraints. Any publication claim must ultimately be supported by a clear comparison with the closest prior work, reproducible evaluation, and measured results.

REFERENCES

- [1] “BD-FakeDetect: A manually curated Bangla dataset for misinformation detection,” 2024. [Details to be verified against the original publication before final submission.]
- [2] F. T. J. Faria et al., “MultiBanFakeDetect: Integrating Advanced Fusion Techniques for Multimodal Detection of Bangla Fake News in Under-Resourced Contexts,” *International Journal of Information Management Data Insights*, vol. 5, no. 2, Art. no. 100347, 2025.
- [3] “BTMD: A Bengali trimodal dataset for misinformation detection,” 2025. [Details to be verified against the original publication before final submission.]
- [4] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, “FEVER: A Large-scale Dataset for Fact Extraction and VERification,” in *Proc. NAACL-HLT*, 2018, pp. 809–819.
- [5] J. Schlichtkrull et al., “AVeriTeC: A Dataset for Real-world Claim Verification with Evidence from the Web,” in *Proc. NeurIPS Datasets and Benchmarks Track*, 2023.
- [6] B. M. Yao, A. Shah, L. Sun, J.-H. Cho, and L. Huang, “End-to-End Multimodal Fact-Checking and Explanation Generation: A Challenging Dataset and Models,” in *Proc. ACM SIGIR*, 2023, pp. 2733–2743.
- [7] S.-I. Papadopoulos, C. Koutlis, S. Papadopoulos, and I. Kompatsiaris, “VERITE: A Robust Benchmark for Multimodal Misinformation Detection Accounting for Unimodal Bias,” *arXiv preprint arXiv:2304.14133*, 2023.
- [8] M. Chakraborty et al., “FACTIFY3M: A Benchmark for Multimodal Fact Verification with Explainability through 5W Question-Answering,” *arXiv preprint arXiv:2306.05523*, 2023.
- [9] “EVID-Bench: A Benchmark for Search-Grounded Video Misinformation Detection,” 2026. [Preprint; details to be verified.]
- [10] “Micro-video Misinformation Detection with External Evidence Retrieval,” 2026. [Details to be verified against the original publication before final submission.]